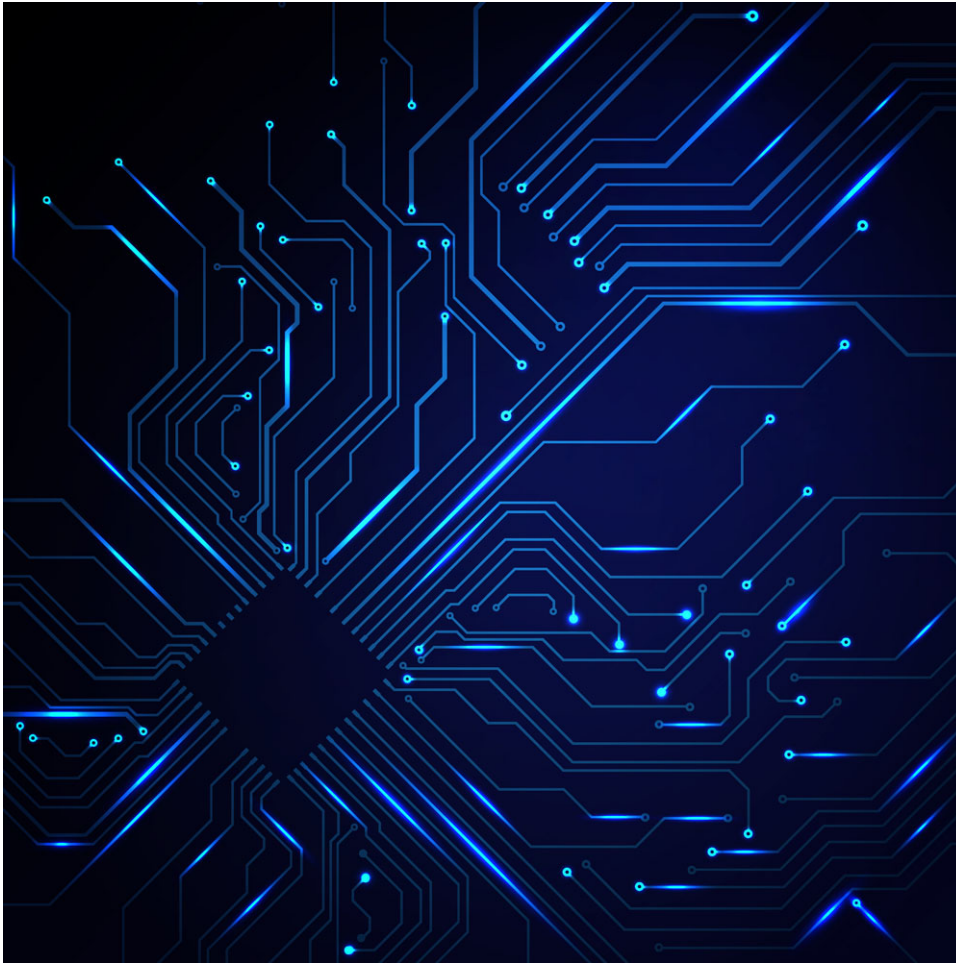




LAB 4 BASIC LOGIC GATE IMPLEMENTATION

Group : 14

Name : 黃翊宏

Student ID : 01357101

Note

1. All files must be converted to PDF format before submission.
2. **Submission deadline:**
Next Tuesday at 1:00 PM
3. Each person submits one assignment.

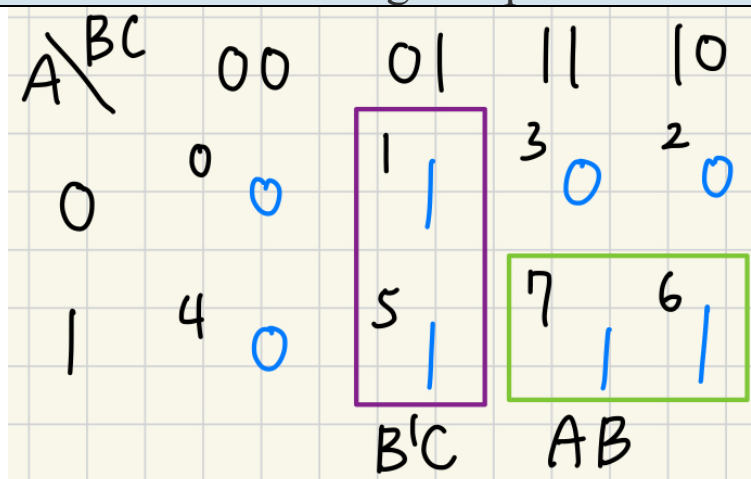
Lab 4-1

Truth Table (Voltage)			
<i>A</i>	<i>B</i>	<i>C</i>	<i>F</i>
0	0	0	0.21
0	0	1	2.71
0	1	0	0.21
0	1	1	0.21
1	0	0	0.21
1	0	1	2.71
1	1	0	2.71
1	1	1	2.71

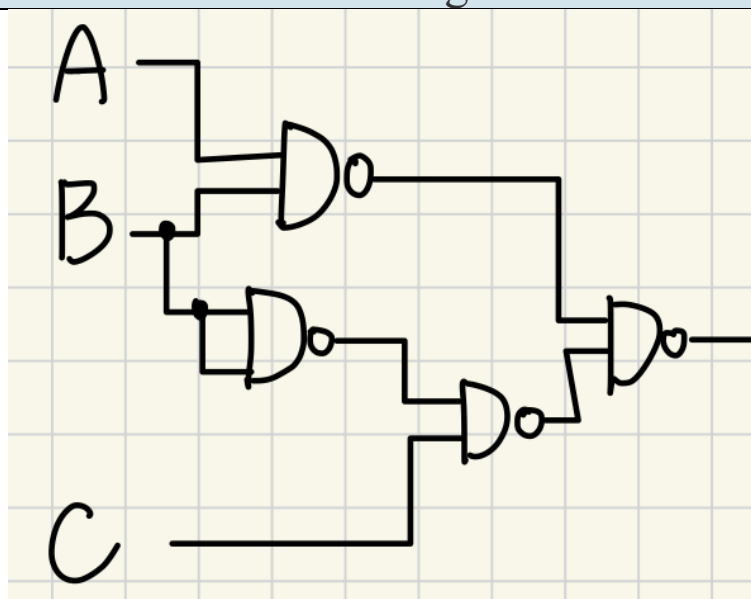
Minimal Boolean
Expression

$$AB + B'C$$

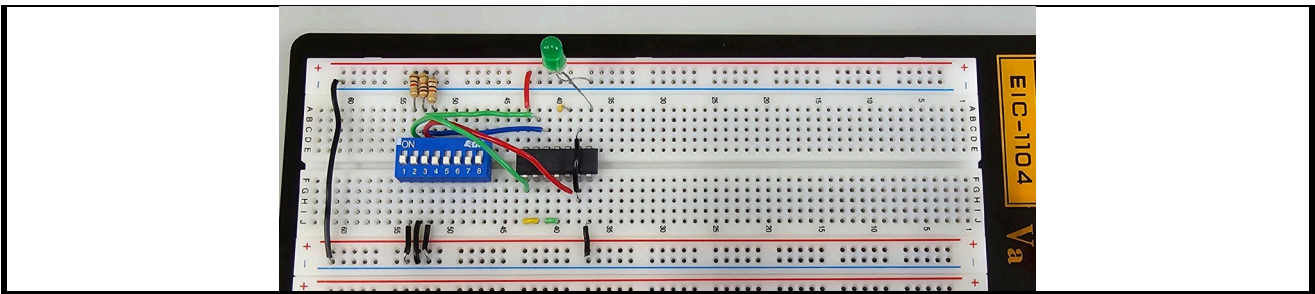
Karnaugh map



Circuit Diagram



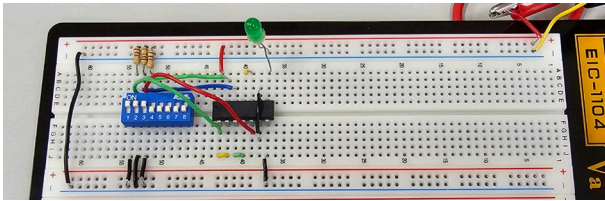
Breadboard Wiring Photo



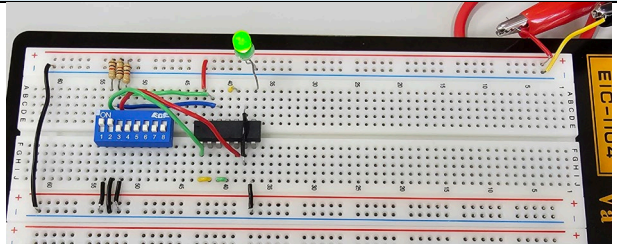
Breadboard Wiring Photos

(Please neatly connect the solid-core wires to the breadboard)

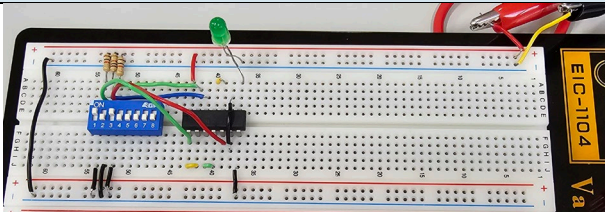
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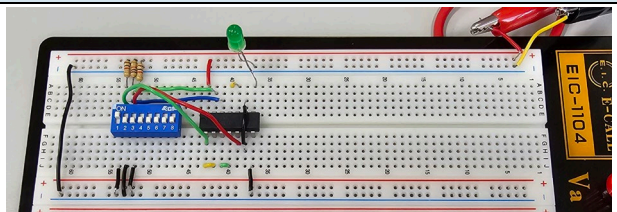
$$A = 0, B = 0, C = 1$$



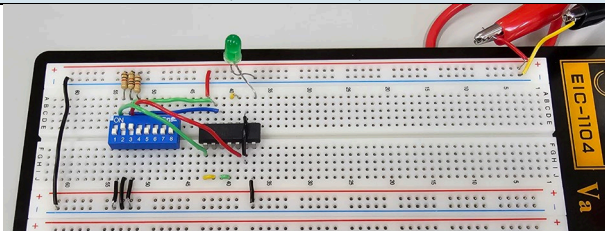
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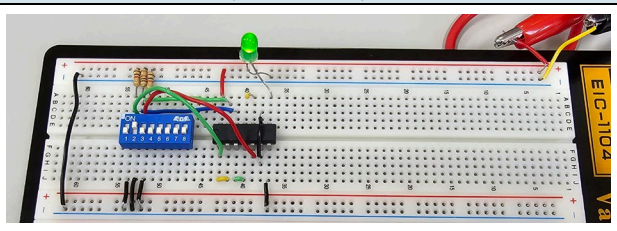
$$A = 0, B = 1, C = 1$$



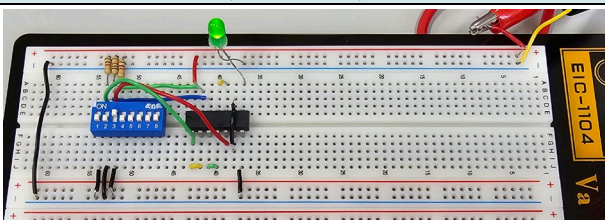
$$A = 1, B = 0, C = 0$$



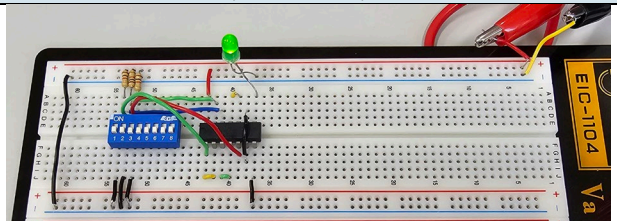
$$A = 1, B = 0, C = 1$$



$$A = 1, B = 1, C = 0$$

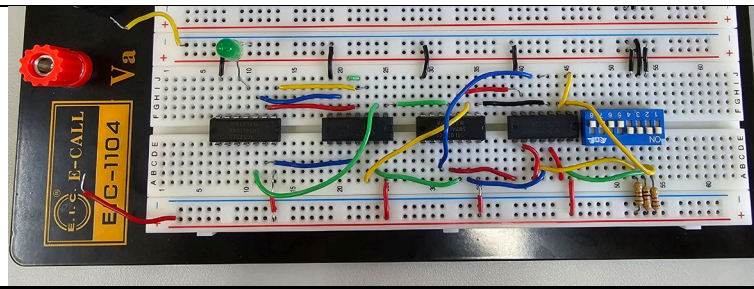


$$A = 1, B = 1, C = 1$$



★ Lab 4-2

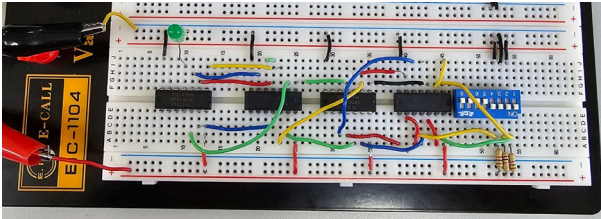
Truth Table (Voltage)			
A	B	C	F
0	0	0	0.21
0	0	1	2.71
0	1	0	0.21
0	1	1	0.21
1	0	0	0.21
1	0	1	2.71
1	1	0	2.71
1	1	1	2.71
Minimal Boolean Expression	$A'B + B'C'$		
Karnaugh map			
Circuit Diagram			
Breadboard Wiring Photos			



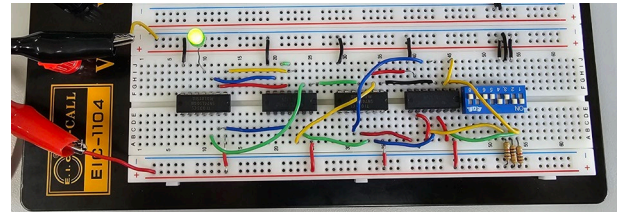
Breadboard Wiring Photos

(Please neatly connect the solid-core wires to the breadboard)

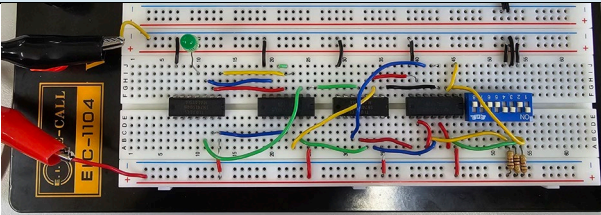
$$A = 0, B = 0, C = 0$$



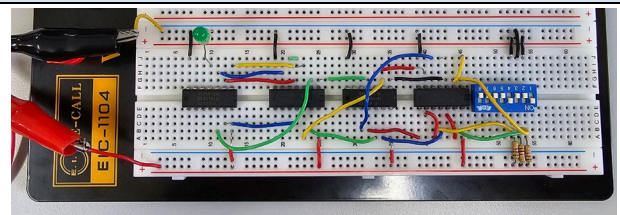
$$A = 0, B = 0, C = 1$$



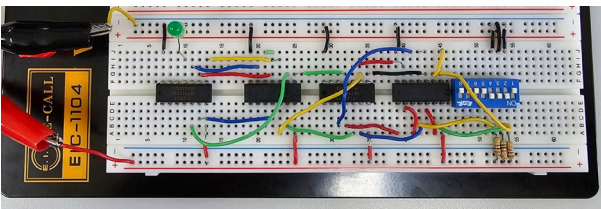
$$A = 0, B = 1, C = 0$$



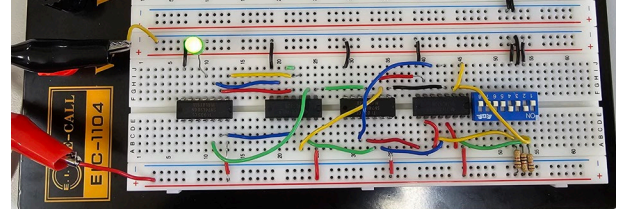
$$A = 0, B = 1, C = 1$$



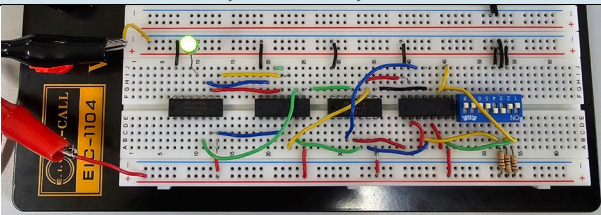
$$A = 1, B = 0, C = 0$$



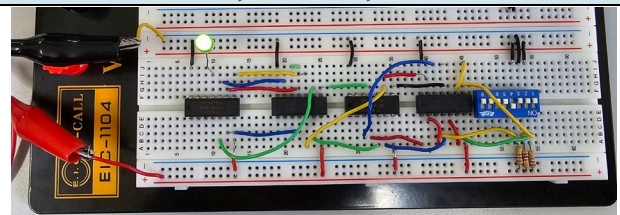
$$A = 1, B = 0, C = 1$$



$$A = 1, B = 1, C = 0$$



$$A = 1, B = 1, C = 1$$



★ Lab 4-3

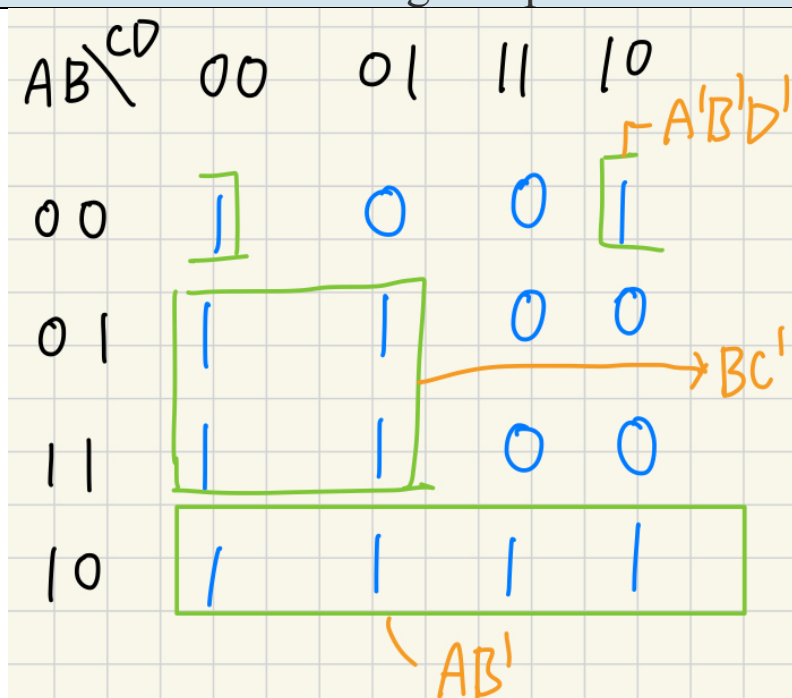
Truth Table (Voltage)

A	B	C	D	F
0	0	0	0	2.71
0	0	0	1	0.21
0	0	1	0	2.71
0	0	1	1	0.21
0	1	0	0	2.71
0	1	0	1	2.71
0	1	1	0	0.21
0	1	1	1	0.21
1	0	0	0	2.71
1	0	0	1	2.71
1	0	1	0	2.71
1	0	1	1	2.71
1	1	0	0	2.71
1	1	0	1	2.71
1	1	1	0	0.21
1	1	1	1	0.21

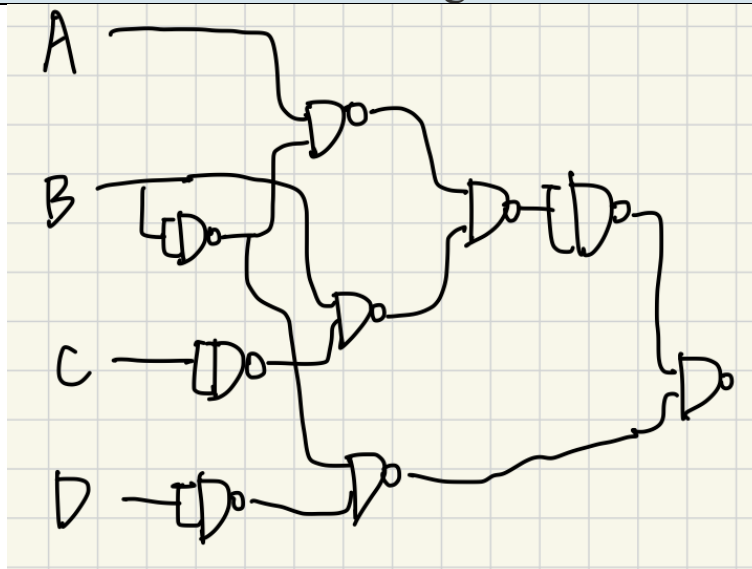
Minimal
Boolean
Expression

$$AB' + BC' + A'B'D'$$

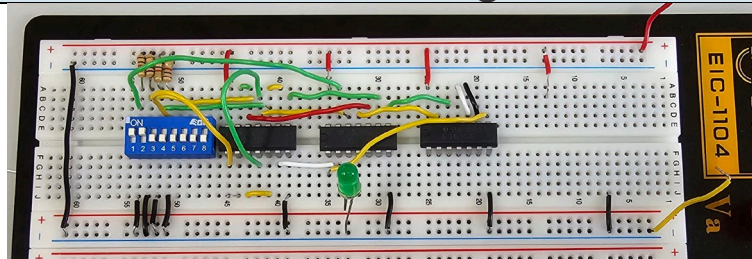
Karnaugh map



Circuit Diagram



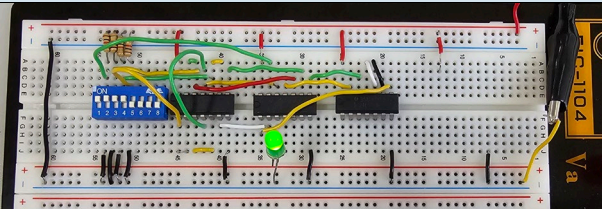
Breadboard Wiring Photos



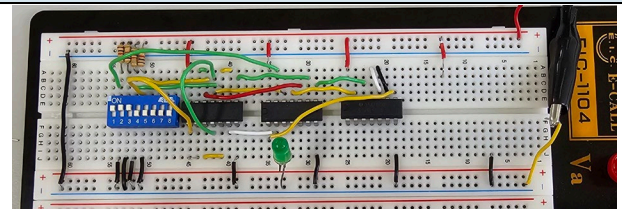
Breadboard Wiring Photos

(Please neatly connect the solid-core wires to the breadboard)

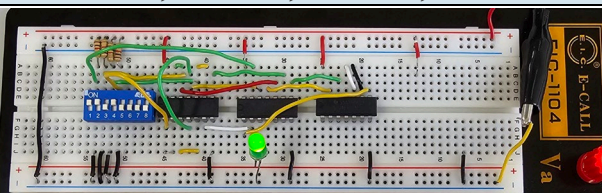
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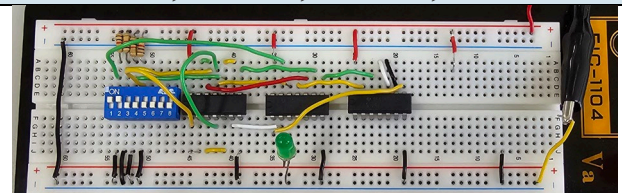
$A = 0, B = 0, C = 0, D = 1$



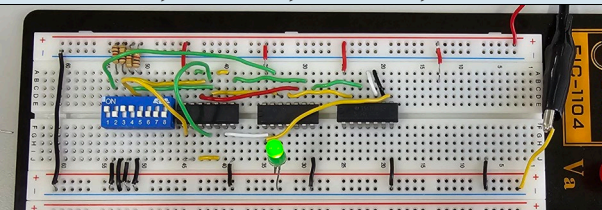
$A = 0, B = 0, C = 1, D = 0$



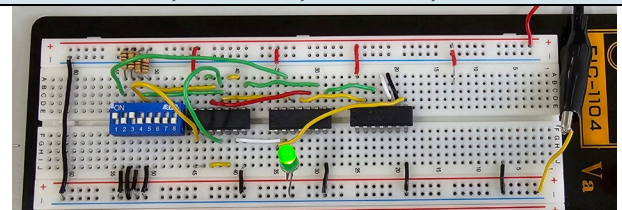
$A = 0, B = 0, C = 1, D = 1$



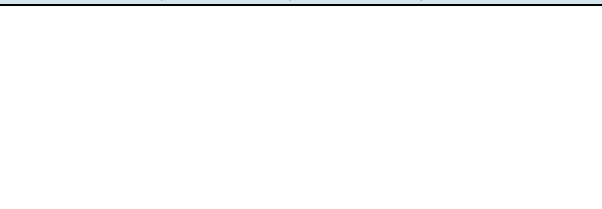
$A = 0, B = 1, C = 0, D = 0$



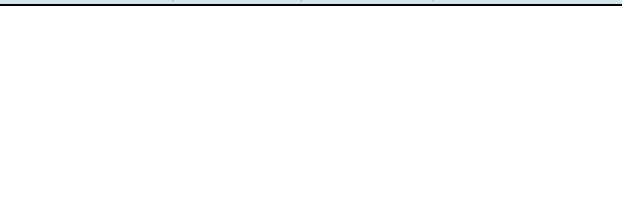
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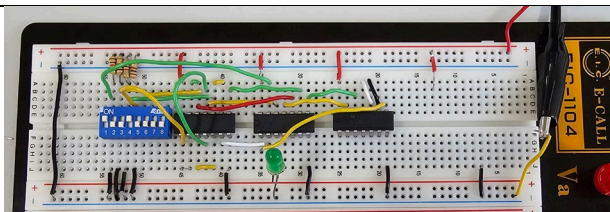


$A = 0, B = 1, C = 1, D = 0$

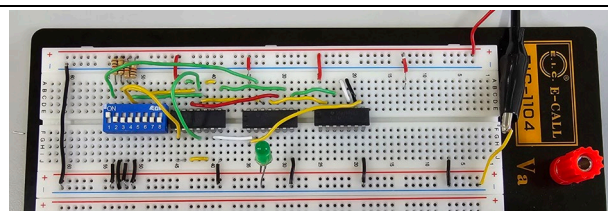


$A = 0, B = 1, C = 1, D = 1$

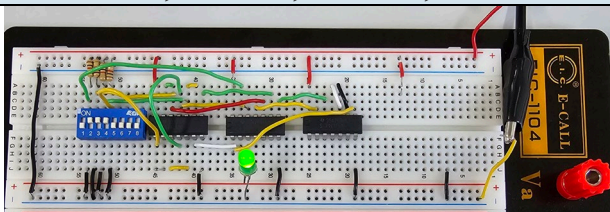




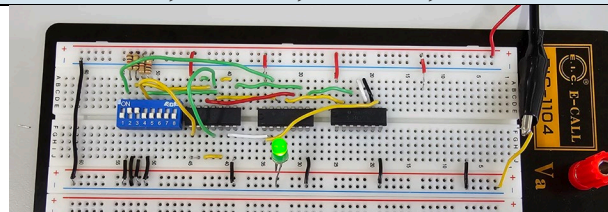
$$A = 1, B = 0, C = 0, D = 0$$



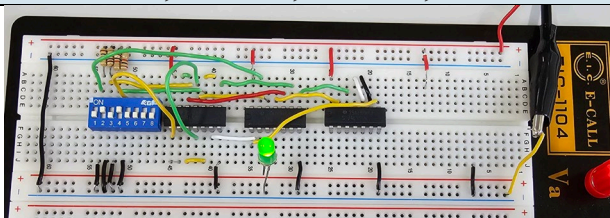
$$A = 1, B = 0, C = 0, D = 1$$



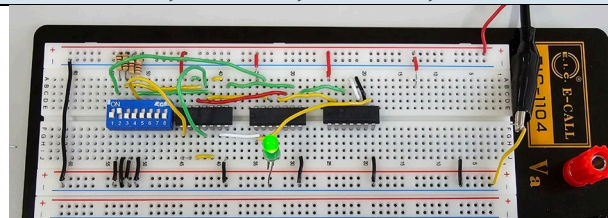
$$A = 1, B = 0, C = 1, D = 0$$



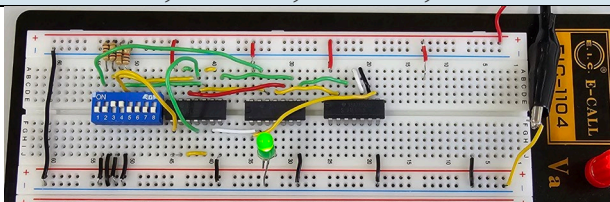
$$A = 1, B = 0, C = 1, D = 1$$



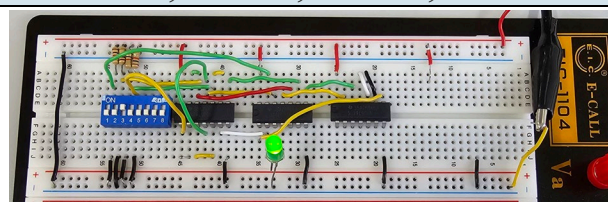
$$A = 1, B = 1, C = 0, D = 0$$



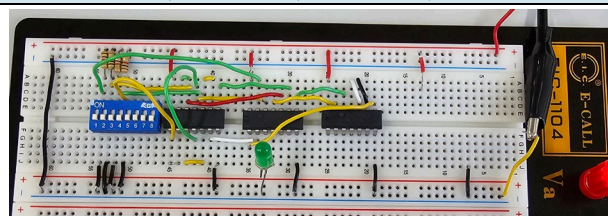
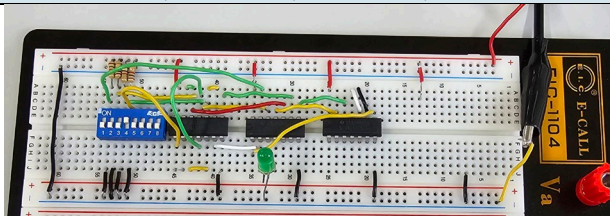
$$A = 1, B = 1, C = 0, D = 1$$



$$A = 1, B = 1, C = 1, D = 0$$



$$A = 1, B = 1, C = 1, D = 1$$



Lab 4-4

Truth Table (Voltage)

A	B	C	D	F
0	0	0	0	2.71
0	0	0	1	0.21
0	0	1	0	2.71
0	0	1	1	0.21
0	1	0	0	2.71
0	1	0	1	2.71
0	1	1	0	0.21
0	1	1	1	0.21
1	0	0	0	2.71
1	0	0	1	2.71
1	0	1	0	2.71
1	0	1	1	2.71
1	1	0	0	2.71
1	1	0	1	2.71
1	1	1	0	0.21
1	1	1	1	0.21

Minimal
Boolean
Expression

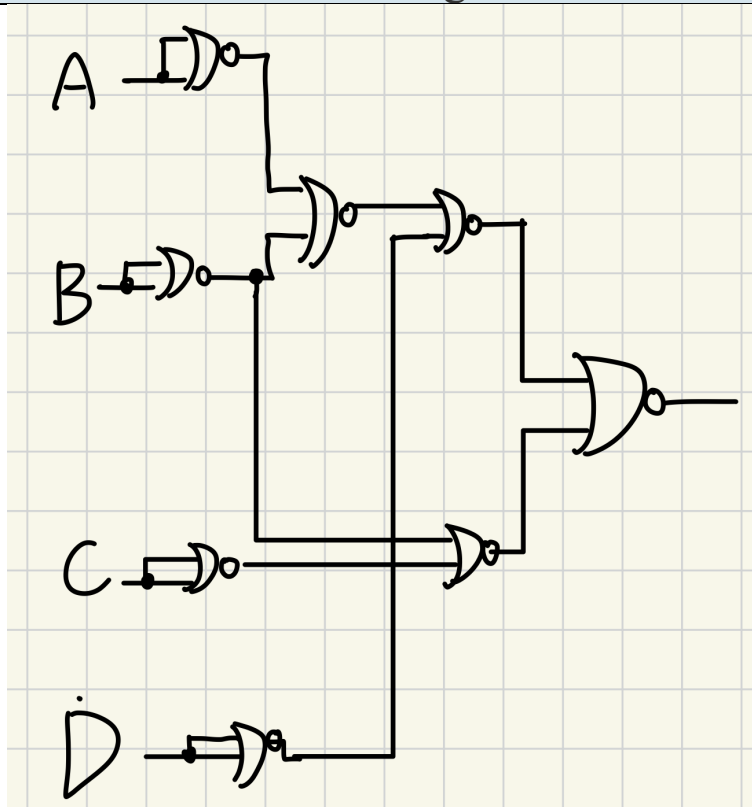
$$A'B'D + B'C'$$

Karnaugh map

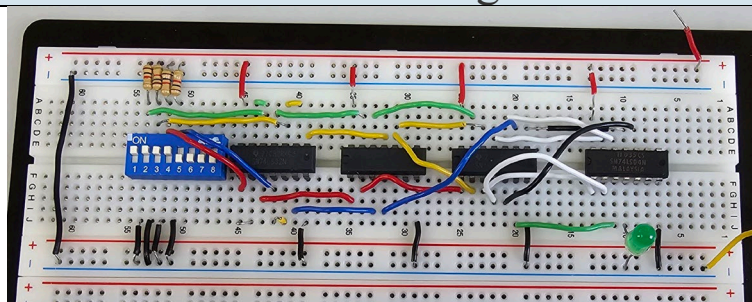
$AB \backslash CD$	00	01	11	10
00		0	0	
01			0	0
11			0	0
10				

$A'B'D$ (grouping 00, 01)
 $B'C'$ (grouping 11, 10)

Circuit Diagram



Breadboard Wiring Photos

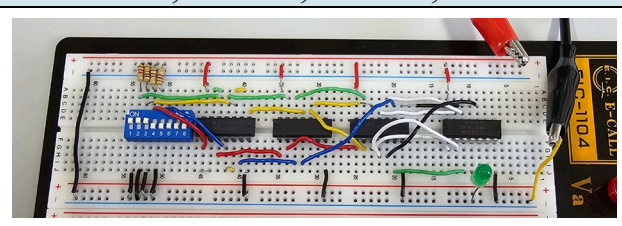
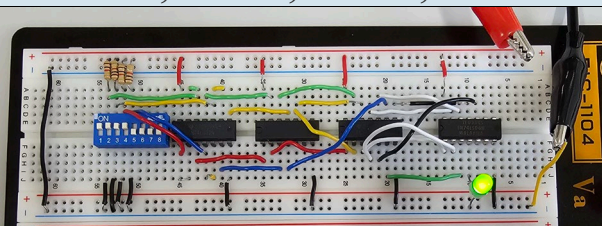


Breadboard Wiring Photos

(Please neatly connect the solid-core wires to the breadboard)

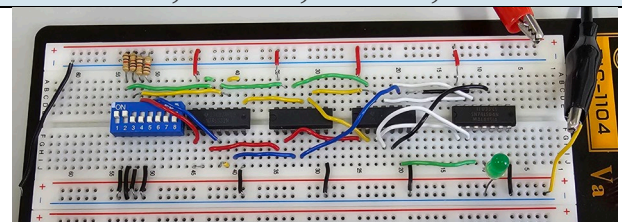
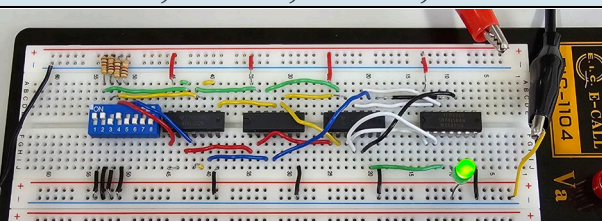
$A = 0, B = 0, C = 0, D = 0$

$A = 0, B = 0, C = 0, D = 1$



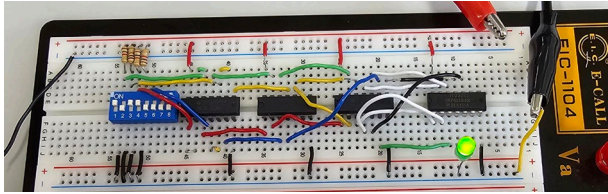
$A = 0, B = 0, C = 1, D = 0$

$A = 0, B = 0, C = 1, D = 1$

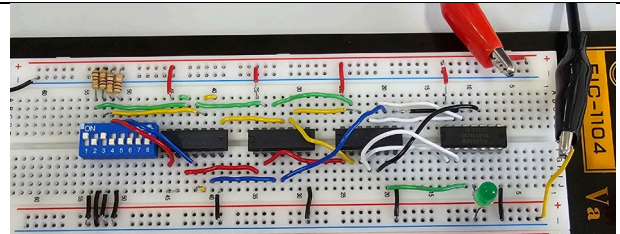


$A = 0, B = 1, C = 0, D = 0$

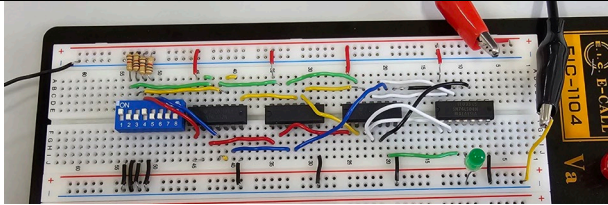
$A = 0, B = 1, C = 0, D = 1$



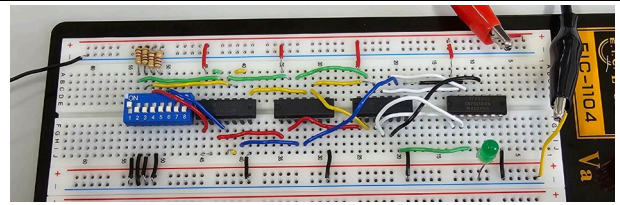
$$A = 0, B = 1, C = 1, D = 0$$



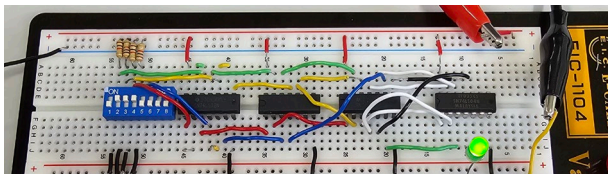
$$A = 0, B = 1, C = 1, D = 1$$



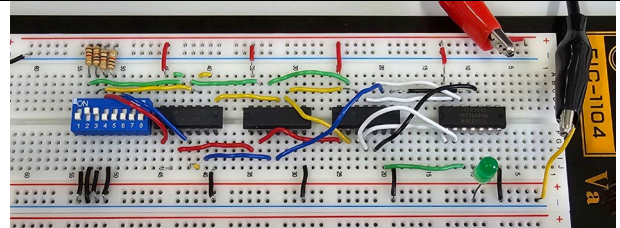
$$A = 1, B = 0, C = 0, D = 0$$



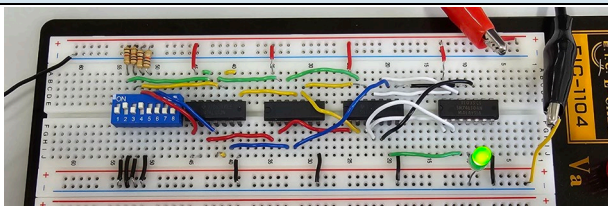
$$A = 1, B = 0, C = 0, D = 1$$



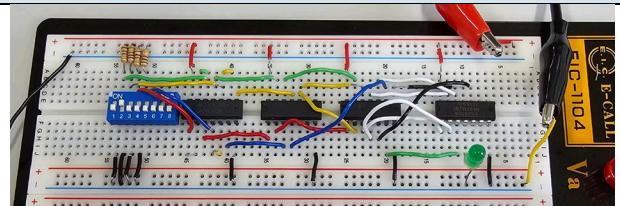
$$A = 1, B = 0, C = 1, D = 0$$



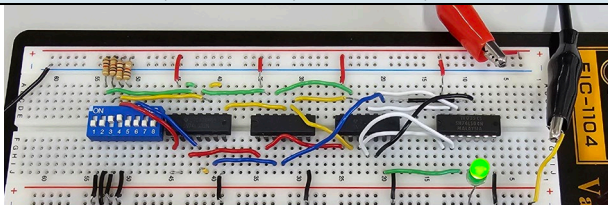
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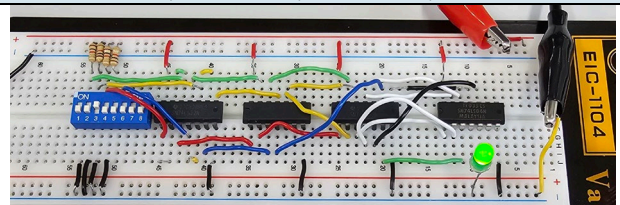
$$A = 1, B = 1, C = 0, D = 0$$



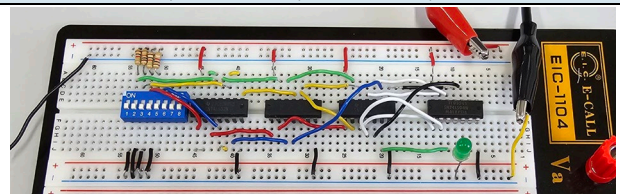
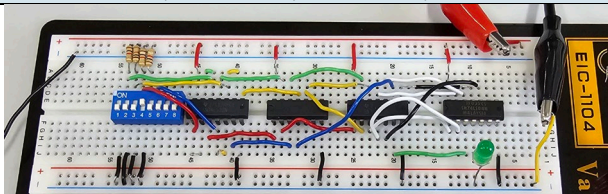
$$A = 1, B = 1, C = 0, D = 1$$



$$A = 1, B = 1, C = 1, D = 0$$



$$A = 1, B = 1, C = 1, D = 1$$



Lab Summary and Reflection :

Summary:

In this lab experiment, we explored the practical application of digital logic design by simplifying Boolean expressions and implementing logic circuits using a limited set of logic gates. The primary tasks involved utilizing Karnaugh maps to reduce Boolean functions to their minimal forms, as well as applying the sum-of-products and product-

of-sums techniques learned in the previous class to manage the Boolean expressions effectively. From the simplified expressions, we developed circuit diagrams and built the corresponding circuits on a breadboard, strictly using the specified logic gates such as two-input NAND or NOR gates. Comprehensive truth tables were constructed to verify the voltage levels—where approximately 0.21 V represented a logic low and 2.71 V indicated a logic high—and the functionality of the circuits was confirmed by testing each input pattern and observing the LED outputs. This experiment seamlessly integrated theoretical Boolean algebra with hands-on circuit design and practical implementation.

Reflection:

This laboratory exercise provided a valuable opportunity to bridge the gap between abstract logic concepts and tangible hardware implementation. Working through the process of minimizing Boolean expressions using Karnaugh maps deepened my understanding of logical optimization, and applying both the sum-of-products and product-of-sums methods underscored the importance of flexible problem-solving strategies in digital design. The constraints of using only a limited set of logic gates challenged me to think creatively and precisely when translating theory into practice, particularly during the circuit design and breadboard assembly phases. Despite encountering initial challenges with ensuring accurate voltage levels and secure wiring connections, the iterative troubleshooting process was instrumental in enhancing my practical skills and reinforcing the critical principles of digital logic design. Overall, the lab not only bolstered my grasp of Boolean simplification techniques and circuit implementation but also underscored the significance of careful planning and methodical execution in engineering projects.